

CHAPTER 6

DIABETIC RETINAL TWO STAGE INTERPRETABLE GLAUCOMA INTELLIGENCE SYSTEM

6.1 OVERVIEW OF PROPOSED RESEARCH

Researchers have been working on different perception to perceive the glaucoma. They used numerous techniques to extract the ailment from eye fundus. Each technique has some pitfalls. So, this cram noted the issues and listed below,

- Trustworthy predictions required on diabetic patients'
- Manual selection required for choosing the best and strong classifier
- Mostly Feature extraction methods based on discrete wavelet transform and high order spectra of digital signalling. Computationally intensive one.
- Lacking of strong learners in training phase
- Two phase predictions require to wipe out blindness
- Sustain to any processors
- Space efficiency

To trounce the above issues, the novel methodology is proposed to predict the diabetic retinal glaucoma from the diabetic patient's general medical record and upright confirmation on predicted patients 2D fundus.

6.2 PROPOSED METHODOLOGY

First and foremost, explainable AI used to predict retinal disease on diabetic patient's medical records. After the above model, the expected patient's retinal fundus images are preprocessed and segmented. Then the statistical methods are used for extracting the retinal special features of an image. The average, variance of a single pixel value, and spatial pixels in an image are the first order statistical features. The Grey Level Co-occurrence Matrix (GLCM), which estimates the values between two pixels, is used to extract the second order

texture attributes. The Grey Level Run Length Matrix (GLRLM) is determined the relationship between more than two pixels in an input image in order to extract higher order statistical texture information. ML ensemble classification techniques are fed an extracted resultant feature matrix. K-fold cross validation routine is used for training ensemble classifier. Real time dataset is used for evaluating the performance of the improvised an interpretable explainable AI ensemble prediction model (EAIHSEC) (Figure 6.1).

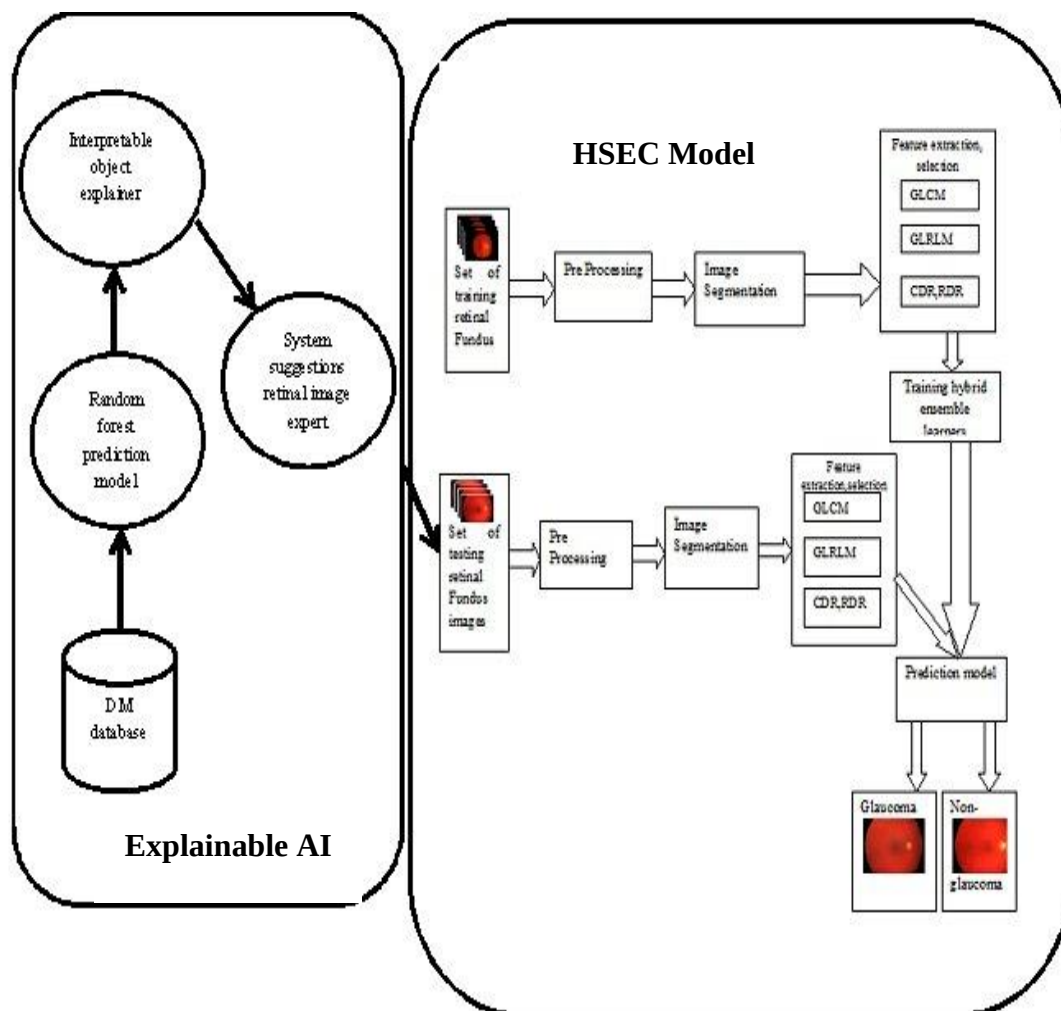


Figure 6.1 Proposed Explainable Artificial Intelligence Hybrid Stacking Ensemble Classification Model

6.2.1 Interpretable Explainable AI

Local Interpretable Model Agnostic Explanations (LIME) is used to interpret and find the local fidelity of diabetic parameters prediction. This is persistent to agnostic to all representation of diabetic dataset. Foremost, diabetic mellitus people dataset is pre-processed by imputation and make it as a valid set for further processing. In a second phase, legitimate dataset is fragmented in to training and testing. Next, different classifiers applied and based on performance analysis best fit model is random forest. In a fourth step, fit model is trained through interpretability explainer object to make new predictions on diabetic features. The set of feature selection is based on maximum weight of ridge and Gower's distance for apiece of permutations explanation among parameters to predict chance and no chance of retinal glaucoma. This acts as a clinical decision system to avoid blindness on diabetic peoples and message intimation to experts to acquire fundus image examination.

6.2.2 Preprocessing and Segmentation

Preprocessing is the primary step to increase the superiority of the images for segmentation, object detection and classification processing. This process is used to enhance some features and remove some unwanted features from an image to make effective one for easy diagnosis. An unprocessed input image is standardizing in the array $[0, 1]$. Colour fundus image is transformed in to gray scale image for further processing. Gaussian filters are used for removing the existing noise. Then histogram equalization is employed to expand the contrast of the retina.

6.2.3 Statistical Texture Feature Extraction through GLCM and GLRLM

One imperative phase of medical picture scrutiny is the extraction of texture features. The segmented image continues this procedure. Histogram equalisation, local binary pattern, grey level, registration, and autocorrelation techniques are among the various approaches used to extract features. Co-occurrence matrices are employed in this suggested technique to extract

characteristics from the processed image. GLCM and GLRLM are employing to extract the pixel values of first order, second order, and high order features from this image (Figure 6.2).

The grey values of i , distributed in the spatial relationship of the specific offset pixel value j at given distance d and θ angle, reflect this co-occurrence distribution matrix. In the co-occurrence matrix representation, the first pixel is now referred to as the reference pixel, while the other pixel is known as the neighbour pixel. When the input picture I is analysed and has a dimension of $M \times M$, the co-occurrence matrix P is shown as

$$P(i, j) = \sum_{x=1}^M \sum_{y=1}^M = \begin{cases} 1, & \text{if } I(x, y) = i \text{ and } I(x + \Delta x, y + \Delta y) = j \\ 0, & \text{Otherwise} \end{cases} \quad (6.1)$$

Here $(\Delta x, \Delta y)$ is offset and represents the interest of pixel distance of the neighbours. The algorithm steps are elucidated below,

1. Get the pre-processed image, I
2. Read the data of the I
3. Create co-occurrence matrix
4. Feature extraction
5. Acquired features are written into csv data file

Based on the above processing GLCM features are segregated from the segmented fundus images. GLRL Matrix is a run length matrix, which holds the values of gray level (gl), run length (rl) and direction (θ) of the fundus image. These are searching towards the specified direction and same gray level value of the pre-processed retinal image. This measures the consecutive pixel values for the same gray value for selected direction.

$$S(i, j) = (g(i, j) | i), 0 \leq i \leq Mg, 0 \leq j \leq Smax, \quad (6.2)$$

where Mg is the maximum gray level of an image and $Smax$ is the maximum length.

These features are extracted through the following steps with appropriate mathematical calculation and output feature values stored in the data file.

1. Acquire an image, I
2. Pixel selection for passing (d, θ)
3. Matrix Construction in S
4. Replacing the position of the pixels, if selected and consecutive pixels are same
5. Repeat the process until features extracted
6. Save the results in the data file

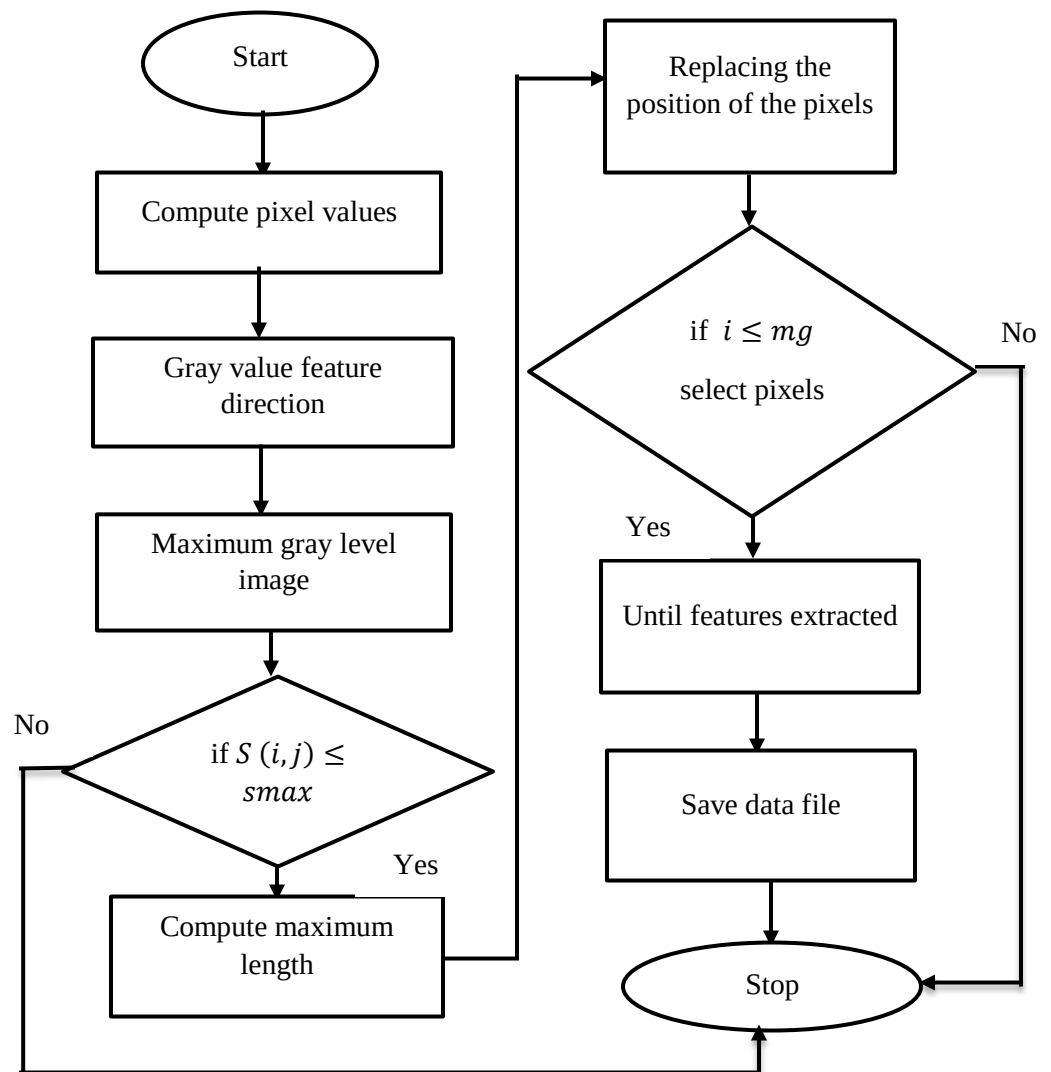


Figure 6.2 Flowchart Diagram of GLCM and GLRLM

6.2.4 Glaucoma Detection Factors Based on Entropy and Kurtosis Statistical Features

Generally, glaucoma deduction factors are Cup to Disc Ratio (CDR) and Rim to Disc Ratio (RDR). These features remains calculated through the segmented FR image. Statistical entropy and kurtosis are depending on the gray values of the original FR image. Cup region entropy calculated from green channel. The OD and OC are deducted from the rim region entropy value. Entropy follows the restoration of the original values from the green channel.

Another common statistical feature kurtosis is used to measure the cup region distribution probability of the fourth moment of the segmented input fundus. Lower kurtosis is good for further process. It is calculated by the following, $k = ((a_i - \bar{a})/I)/S^4$. Here, $\bar{a}$ is the mean, S is standard deviation, and I is the size of the sample image.

6.3 ENSEMBLE LEARNING CLASSIFIER

After feature extraction, eighteen features are selected for further glaucoma classification. Now, existing data are separated into learning set and examination set. Learning set enclosed with one class label which is acting as a target instance with many features (attributes). In this proposed method, the labels are known to the model. So, the supervised ML classifiers are coming in to the processing of glaucoma prediction. This system is implemented in an ensemble learning classifiers for improving the performance of the classification.

Ensemble approaches are incorporating many methods in to one classification model. It is minimizing the uncertainty through bagging, boosting, or maximizing the predictions as stacking.

6.3.1 Boosting

Boosting is a short tree representation with single decision point. These decision trees are called as Decision Stumps (DS). Training set is weighted and

each weight is revised stand on overall accuracy of the classifiers. Successive models are instructed and included until reaching the minimum accuracy. Every model is weighted based on its expertise and joined together the all-predictions model for testing the new data (Figure 6.3). Adaboost is a famous algorithm, which holds several numbers of meta classifiers for classification. An elucidation of algorithm steps is given below,

Input:

- a. Retinal statistical dataset R
- b. Number of learning iteration I and base classifier B as input

Output: Final ensemble model $H(x)$ model prediction based on voting

Initialization: equal data weight distribution to all instances

$$W_n = 1/M, n = 1, 2 \dots m$$

Process:

for i = 1 to I do

$h_i = B(R, R_i)$ train a base learner h_i of dataset

R and sample weight W_i

$$\text{Error computation of } h_i \leftarrow \text{err}_i = \frac{\sum_{n=1}^m W_n T(h_i(X_n) \neq c_n)}{\sum_{n=1}^m W_n}$$

$$\text{Weight evaluation of } h_i \leftarrow \alpha_i = \frac{\log(1 - \text{err}_i)}{\text{err}_i}$$

Set weight

$$W_n = W_n \cdot \exp[\alpha_i T(h_i(x_n) \neq c_n)] \alpha_i = \log(1 - \text{err}_i) / \text{err}_i$$

End

final model $H(x) \leftarrow \text{sign} \sum_{i=1}^I \alpha_i h_i(x)$

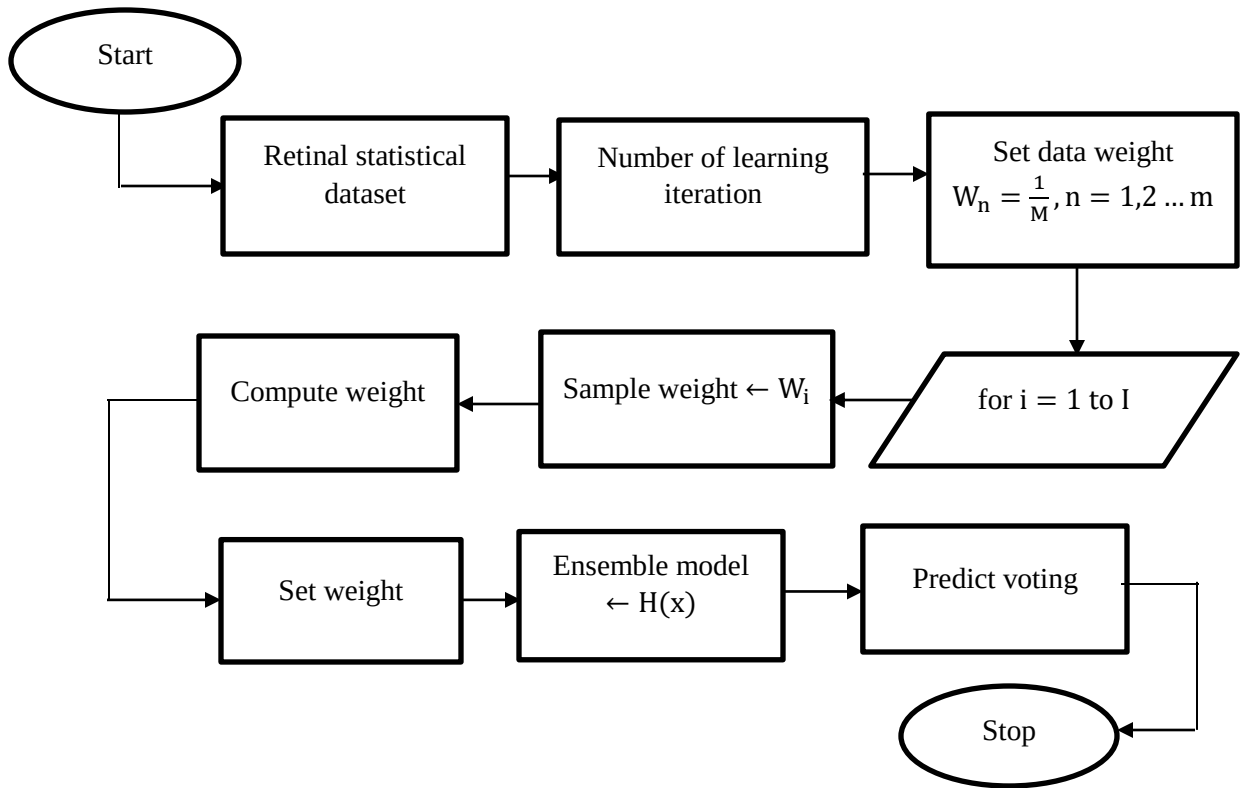


Figure 6.3 Boosting Flowchart Diagram

6.3.2 Hybrid Stacking Ensemble Classifiers (HSEC)

At this juncture, different ML weak classifiers are combined to formulate a new hybrid ensemble learner. In this proposed method, collections of heterogeneous weak ensemble learners are used for final model depiction. Here naïve bayes, SMO of SVM, kNN, and Multi-Layer Perceptron (MLP) are used as weak learners to create an ensemble learner with 10-fold cross validation with hyper parameter tuning. The obtained result is feasible to predict the glaucoma and non-glaucoma (Figure 6.4). This novel HSEC algorithm is explicated below,

Input: Retinal training dataset $R = \{x_i, y_i\}$

Process:

Pace 1: Take up a cross validation to organize second level classifier training set
Indiscriminate split of R in to K equal size subsets; $R = \{R_1, R_2, \dots, R_K\}$

For $k = 1$ to k do //k- different ensemble learners

1.1 Learn first level classifiers

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For t = 1 to T do    // T is first level classifiers
    Learn  $h_{kt}$  classifier from  $R/R_k$  //R is dataset
End

```

1.2 Build a training set for second level classifier

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For  $x_i \in R_k$  do
    Get  $\{X'_i, y_i\}$ , Where //  $X'_i$  is feature vector, h is second level classifier
     $X'_i = \{h_{k1}(x_i), h_{k2}(x_i), \dots, h_{kT}(x_i)\}$ 
End
End

```

Pace 2: learning of second level classifier

gain knowledge of a new classifier 'h' from the listing of $\{X'_i, y_i\}$

Pace 3: first level classifier relearning

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For t ← 1 to T do

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    Learn  $h_t$  on R

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End

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Return  $H(x) = h'(h_1(x), h_2(x), \dots, h_T(x))$  // H(x) is hybrid classifier

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Output: H as hybrid ensemble classifier

In this proposed system, the performance of Adaboost with various meta classifiers is compared with new HSEC algorithm. Results are evaluated by different metrics to find the best accuracy.

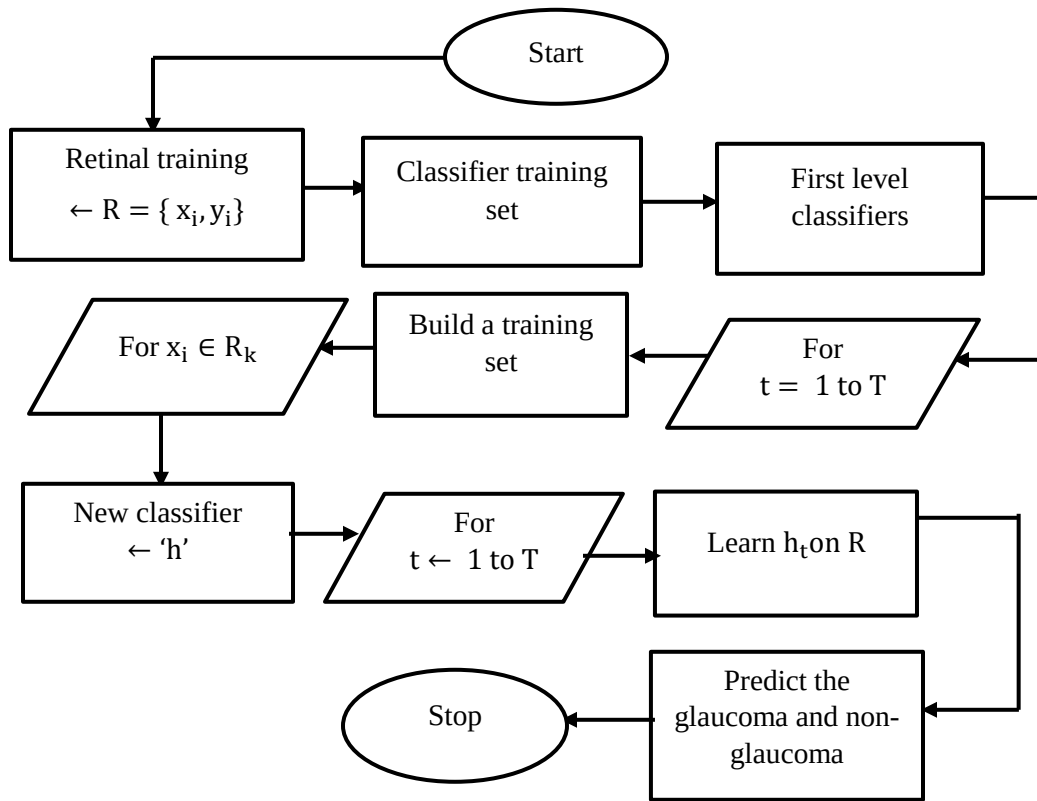


Figure 6.4 HSEC Flowchart Diagram

6.4 RESULTS AND DISCUSSION

This work is put forward to proficient way of glaucoma detection at premature stage, to minimize the optic nerve damage. Student's t test is used for evaluating the following image segmented features CDR and RDR. Results exists statistically remarkable of $p < 0.0001$. After the pre-processed steps, extracted eighteen features are input in to proposed ensemble classifiers along with several meta classifiers to increase the recital (Table 6.1). Proposed algorithms are assessed through the following metrics (Table 6.2).

Segmentation of OD, OC and Rim to Disc Ratio (RDR) are portrayed in earlier. Sample results of the prior one is illustrated in Figure 6.5. Training dataset is an assortment of different public retinal database.

In this projected work, segmented features of OD, OC, RDR and statistical features from GLCM, GLRLM with textural attributes are acquired from the image, and delivered as input for prediction classification. Here, the CDR values of non-glaucoma fundus have from 0.47 ± 0.15 and glaucoma images reclined in the range of 0.60 ± 0.15 . Similarly, the textural features of a healthy image have standard deviation range between 35 ± 125 , and the unhealthy fundus have a more standard deviation value. Entropy ranges between 0.1 and 0.2 of normal image and for glaucoma from 0.3 to 0.4. Another feature, kurtosis of normal image is be positioned between 110 and 180, an abnormal retinal image has the values of 50 to 80.

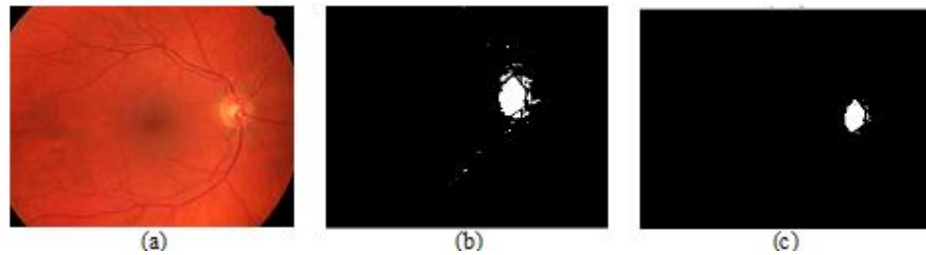


Figure 6.5 Image Segmentation (a) Sample Image (b) OD (c) OC

Other GLCM and GLRLM instances are extracted and altogether created a new dataset with class labeling normal and glaucoma. This set is fed in to first level learner's classification. Here, ensemble classifiers are executed individually as well as with combination of base level classifiers with metaclassifier. Performance of the suggested approach is trained with one datapool and tested with different retinal set. In this case, the level of prediction model is evaluated by using two different test set. In consequence, testing set is input in to HSEC. RMSE is calculated by residuals squaring, calculate residuals averaging's, and find the resultant's square root. Based on this, the proposed system RMSE is comparing to tuning the metaclassifier (Figure 6.6).

Table 6.1 Performance of Hybrid Stacking Ensemble Classifiers

Ensemble Base Classifier	Accuracy	Sensitivity	Specificity	Precision	F1score	RMSE
Naïve Bayes	0.9192	0.9100	0.9286	0.9286	0.9192	0.2672
kNN	0.9141	0.9101	0.9184	0.9192	0.9146	0.2575
SMO	0.9343	0.9208	0.9485	0.9490	0.9347	0.2369
MLP	0.9246	0.9118	0.9381	0.9394	0.9254	0.2474
Proposed (HSEC)	0.9898	0.9900	0.9897	0.9900	0.9892	0.1010

Table 6.2 Testing Data Performance Measures

Proposed Techniques	Accuracy (%)	Sensitivity (%)	Specificity (%)
HSEC	97.2	97.07	95.87
EAIHSEC	98.3	94.56	93.65

New proposed system is achieved the AUC value of 0.91 on ROC. Based on this it is concluded that the prediction model EAIHSEC is a good one. This curve is related to cost-benefit estimation of decision making in particular threshold of a predicted class. This is computed by sensitivity of the class with specificity of the class-1. Subsequent of metrics evaluation, a comparative analysis is conceded to draw the proposed work. This division is presenting the outcome of statistical, structural and surface features of an eye image, which is used to predict the early stage of a retinal glaucoma.

Different meta classifiers are checked with ensemble Adaboost as a base learner and performance measures. In this, RF achieved the highest accuracy of 94.90% with precision of 94.95 %. Second maximum accuracy is 93.3% obtained

by random tree meta classifier with Adaboost. Over all the accuracy of Adaboost with various meta learners is 93.0%.

The performance of hybrid stacking ensembling classification, which is achieved the overall accuracy of 98.98% while the maximum of 99.99%. This hybrid stacking ensembling is fine with 92% for NB, 91% for kNN, 93% for Sequential Minimal Optimization (SMO), and 92% for MLP (Table 6.1) . The results are apparent from the performance measures to fitting for diagnosing the structural changes of retina for glaucoma. RMSE is considering the performance factor of variance reducing. Tuning parameters are employed to attain the lowest percentage of selected classifiers. Finally, the proposed method achieved 0.1010 of RMSE (Figure 6.7). The recommended process is evaluated on HRF and local diabetic patient's image set. Both are obtained the accuracy of 97.2% and 98.3% of respectively (Table 6.2). Hybrid stacking ensemble is performed well to achieve the good accuracy.

In a testing phase diabetic patients valid attributes are provided in to a best fit trained RF model to predict the feasibility of retinal disease under 10 cross validations. Resulted performance is input into an interpretable agnostic explainer as a part of 5 selected features, the distance of manhattan and gower's distance. An output of this explainer is shown the probability of chance, no chance based on weight as explanation fit of 0.100, 0.092 , 0.095 for occurrence of disease in a sample test set (Figure 6.8) . The plot is navigating the correlation among features as support and contradicts. Based on interpretability, an implication is sent to experts to take fundus shot to prevent blindness. Case number 22, 35, and 39 fundus images are taken and provided as input to HSEC system to assure the prediction as chance or not chance of glaucoma.

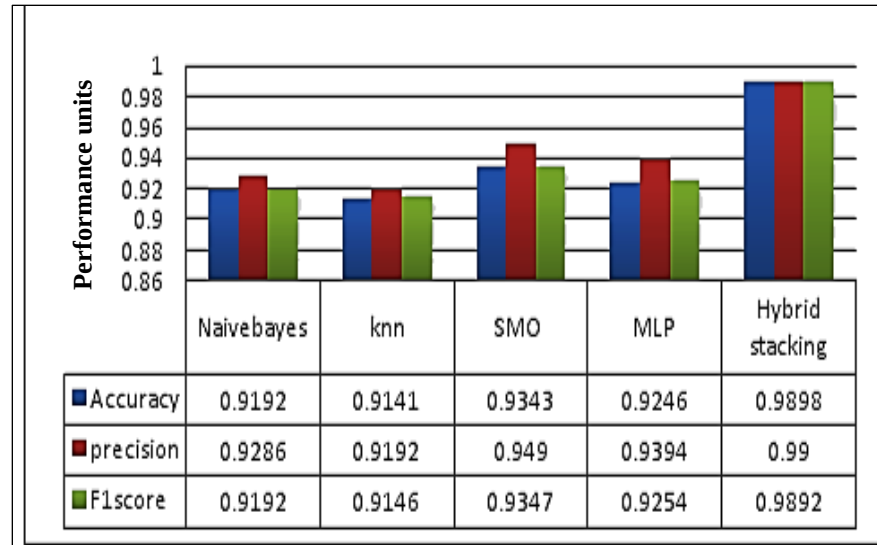


Figure 6.6 Performance Measures of HSEC

This system is resourceful to the succession of glaucoma prediction at the early stage. Most of the existing systems are based on only structural features and some statistical features.

Table 6.3 Assessment with Similar Methods

Authors	Techniques	Classifier	No. of Features	Accuracy (%)
Singh et al. [93]	Wavelet	SVM	18	95.24
Yun et al. [94]	Texture , wavelet	SVM	16	98
Kasu et al. [95]	Dual tree complex wavelet, CDR	MLP	4	97.6
Kishore B and Ananthamoorthy N.P [96]	Feature fusion, classifier fusion	SVM, RF, kNN	fused	99.2
Proposed (EAIHSEC)	GLCM, GLRLM, structure, LIME	HSEC	18	98.9

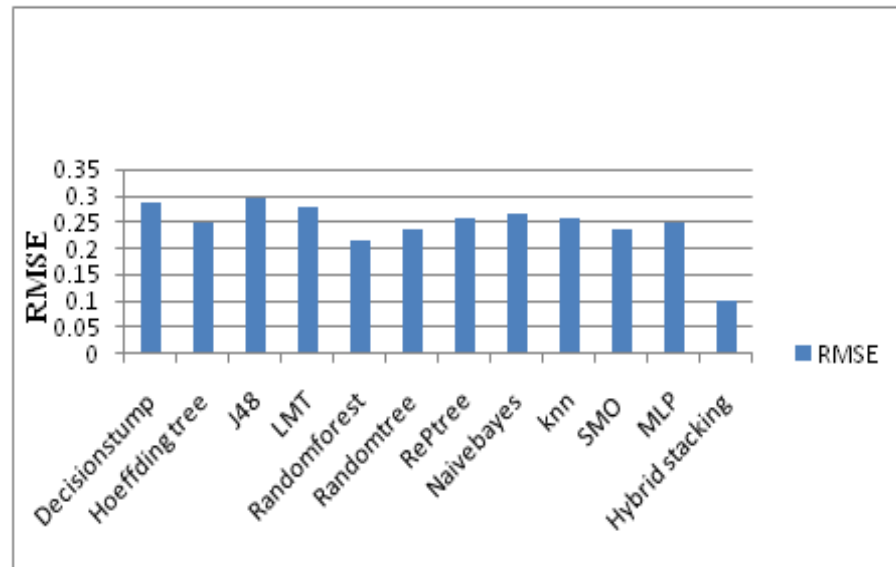


Figure 6.7 RMSE Rate of Ensembles

In this work, the originality lies in the collection of structural, textural and statistical features which are joined together and formulated an input set for hybrid ensemble classification. To boost up the accuracy, weights of the ensembles are adjusted until to reach the best solution. Some of the existing methods are depicted in Table 6.3.

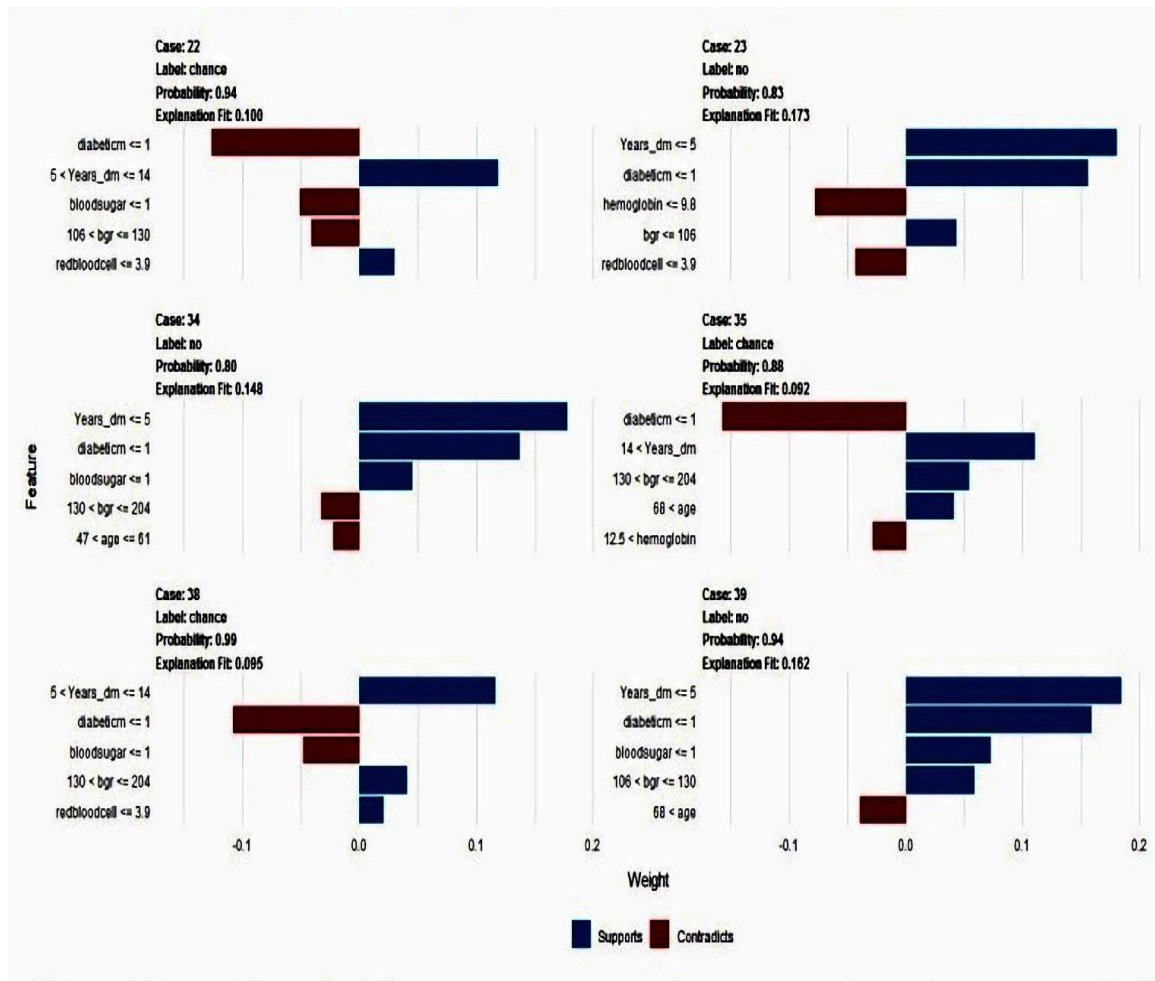


Figure 6.8 Performance of Explainable AI

6.5 SUMMARY

The proposed EAIHSEC model has been tested with two different databases to assure the model efficiency. This examination is diverging from training model, for the reason training and test set both belongs to same retinal database. Using dedicated testing local diabetic patient data set for the above predicted model, with the accuracy of 98.3% and 0.88 of AUC on respectively. This experimental method is screening the accuracy, sensitivity, specificity, precision and less error rate with an enhanced well-suited result. This proposed method can be employed as an assessment support system for diagnosing the glaucoma at premature stage to avoid irretrievable vision.